

# Rahul Rawat

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## PROFILE

Master of Science Data Science student at SRH Heidelberg with hands on experience in artificial intelligence, automation and applied machine learning. I have built generative AI systems with agentic routing and conversational memory, trained machine learning models on real data, and analysed and visualised business data with Python and SQL. I am looking for a working student position where I can help develop intelligent automation and analytics solutions, and EnBW's Digital Capabilities team rethinking customer service with generative AI is exactly the kind of hands on, technology driven environment I want to contribute to.

## EDUCATION

**M.Sc. Data Science and Analytics**, SRH University of Applied Sciences Heidelberg

*Apr 2025 to Present*

**B.Sc. Computer Science**, GL Bajaj Institute of Technology and Management

*2019 to 2023*

## EXPERIENCE

**eRay GmbH**

*Data Scientist · Oct 2025 to Mar 2026*

- Built an end to end recursive time series pipeline forecasting four water quality indicators, chlorophyll a, turbidity, pH and dissolved oxygen, over a six month academic and industry collaboration.
- Benchmarked six models head to head, Ridge, Gradient Boosting, LightGBM, XGBoost, CatBoost and Prophet, and used CatBoost multi quantile regression to deliver asymmetric 80 percent prediction intervals.
- Brought machine learning insight into a real process, surfacing the honest finding that pH and dissolved oxygen are physically predictable while chlorophyll a and turbidity are not without live optical sensors.
- Reconstructed missing winter readings with MICE imputation and wrapped the pipeline in an orchestrator with gate checks that halts on failed imputation rather than letting bad data cascade.

## PERSONAL PROJECTS

**Hybrid RAG Orchestrator: Python, LangChain, Llama 3.1 8b, ChromaDB, HuggingFace, Streamlit**

- Designed a modular generative AI system using Llama 3.1 8b for high speed inference and LangChain for orchestration, the same building blocks behind chatbots and intelligent dialogue systems.
- Engineered a custom decision making router that dynamically classifies user intent into local knowledge retrieval, external web search or direct conversational logic, much like an intelligent assistant deciding how to answer a customer.
- Built a stateful memory agent that maintains conversational context across multiple turns, then shipped the whole system behind a deployed interface with end to end ownership.

### **CreditIQ, Fairness by Design Credit Scoring System: Python, scikit learn, AIF360, SHAP, Streamlit**

- Used machine learning to monitor and improve a decision process, lifting the model's Disparate Impact ratio from a failing 0.79 to a compliant 0.88 and cutting the false negative rate from 44 percent to 16.7 percent.
- Diagnosed a hidden intersectional bias using SHAP, then designed a four way threshold matrix to correct it without over correcting.
- Shipped a Streamlit decision support tool that gives managers a recommendation plus a plain language explanation, keeping a human in the loop.

### **Movie Analytics and Machine Learning Pipeline: GCP BigQuery, Python, SQL, Looker Studio**

- Built an end to end batch pipeline fully automated on a scheduler trigger with no manual steps, an example of the kind of automation that relieves a team of repetitive work.
- Trained a machine learning classifier to predict outcomes before release and surfaced the results in a five page Looker Studio dashboard answering concrete business questions.
- Analysed and visualised key performance indicators for non technical stakeholders, turning raw data into decisions.

## **RESEARCH AND THESIS**

### **Bachelor Thesis, Diabetes Prediction Using Machine Learning: Python, scikit learn, Pandas, Seaborn**

- Built a full end to end machine learning pipeline comparing six classifiers on a clinical dataset of 768 records with 10 fold cross validation and per model confusion matrices.
- Caught biologically impossible zero values the original authors had overlooked, then applied IQR based outlier removal and proper imputation.
- Chose ROC AUC over accuracy as the headline metric for a 65 to 35 imbalanced dataset, avoiding a misleadingly rosy accuracy figure.
- Wrote up the findings as an IEEE style paper including an honest section on limitations and what to do differently.

## **LANGUAGES**

- English: fluent, professional working proficiency
- German: B1, in progress